Time Series Analysis

TSA – DL/ARDL models

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TSA – DL/ARDL models

Linear regression in short

$$\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{\varepsilon}$$

- Y – vector of dependent variable observations (n x 1)
- X – matrix of explanatory variables observations (n x k)
- $\boldsymbol{\beta}$ – vector of parameters (k x 1)
- $\boldsymbol{\varepsilon}$ – vector of error terms (n x 1)

Linear regression -- assumptions

- linear regression model is “linear in parameters.”
- random sampling of observations
- an average error term is equal 0:

$$\boldsymbol{E(\varepsilon|X) = 0}$$

- errors orthogonal to independent variables:

$$\boldsymbol{E(X'\varepsilon|\mathrm{X}) = X'E(\varepsilon|\mathrm{X}) = 0}$$

- constant variance (homoscedasticity) and lack of errors autocorrelation:

$$\boldsymbol{VAR(\varepsilon\varepsilon'|X) = E(\varepsilon\varepsilon'|X) = I\sigma^2}$$

- at least as many observations as variables:

$$\boldsymbol{r(X) = k \leq n}$$

Read: https://www.albert.io/blog/key-assumptions-of-ols-econometrics-review/

OLS estimator

Vector of parameter estimates:

$$\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

Where:

- $\boldsymbol{X}$ – matrix of explanatory variables observations (n x k)
- y – vector of dependent variable observations (n x 1)
- b – vector of parameter estimates (k x 1)

Linear determination coefficient $\boldsymbol{R^2}$

$$\boldsymbol{R^2 = \frac{\sum_{t=1}^{n}\left(\hat{y}_t - \bar{\hat{y}}\right)^2}{\sum_{t=1}^{n}(y_t - \bar{y})^2} = \frac{ESS}{TSS} = (if\ const) = 1 - \frac{RSS}{TSS}}$$

where:

- $\boldsymbol{y_t}$ - value of variable Y at time t
- $\boldsymbol{\hat{y}_t}$ - theoretical (fitted) value of Y at time t
- $\boldsymbol{\bar{y}}$ ($\boldsymbol{\bar{\hat{y}}}$)- average value of variable Y (fitted)

Adjusted linear determination coefficient R^2_{adj}

$$R^2_{adj} = 1 - \frac{n-1}{n-k}\left(1 - R^2\right)$$

where:

- R^2- linear determination coefficient
- n -- number of observations
- k -- number of regressors

OLS linear regression with TSA – DL

Distributed Lags Model:

- Idea: independent variables have an influence on dependent variable in time (not only prompt, but also lagged influence)
  - In the model not only independent variable from the same moment as dependent variable should be included, but also from previous periods
- Example of use: long term Equilibrium process analysis
- DL can be estimated by OLS or GLS if independent variables are strictly exogenous
- DL works well if dependent and independent variables are stationary (otherwise Variance-Covariance matrix estimators are inconsistent, and statistical inference is incorrect, however estimators are consistent)
- DL works well if CLRM assumptions are satisfied (linear functional form, given independent variables, lack of systematic error, homoscedasticity, lack of autocorrelation in residuals)
- DL can works poorly when a lot of lags of independent variable should be used (to eliminate autocorrelation issue; possible solutions ARDL, ARIMA, …)

$$y_t = \mu + \beta_0 x_t + \cdots + \beta_p x_{t-p} + \varepsilon_t$$

Parameter interpretation

- There is little sense to interpret single parameter, rather joint analysis of parameter is interesting
  - Short-term reaction (impact multiplier) of dependent variable on prompt change of independent variable
  - Long-term reaction (long-term multiplier) of dependent variable on independent variables changes

ST

$$E(y_t + \Delta y) = \mu + \beta_0(x_t + \Delta x) + \cdots + \beta_p x_{t-p} + \varepsilon_t$$

$$E(y_t + \Delta y) = \mu + \beta_0(x_t + \Delta x) + \cdots + \beta_p x_{t-p} + \varepsilon_t + \beta_0 \Delta x$$

$$E(y_t + \Delta y_t) = E(y_t) + \beta_0 \Delta x$$

$$\boldsymbol{\frac{E(\Delta y)}{\Delta x} = \beta_0}$$

LT

$$E(y_t + \Delta y) = \mu + \beta_0(x_t + \Delta x) + \cdots + \beta_p\left(x_{t-p} + \Delta x\right) + \varepsilon_t$$

$$E(y_t + \Delta y) = \mu + \beta_0(x_t) + \cdots + \beta_p\left(x_{t-p}\right) + \varepsilon_t + \sum_{i=0}^{p} \beta_i$$

$$E(y_t + \Delta y) = E(y_t) + \Delta x \sum_{i=0}^{p} \beta_i$$

$$\boldsymbol{\beta_{CUM} = \frac{E(\Delta y)}{\Delta x} = \sum_{i=0}^{p} \beta_i} \qquad \boldsymbol{\beta_{LT} = \frac{E(\Delta y)}{\Delta x} = \sum_{i=0}^{\infty} \beta_i}$$

Average reaction lag

$$\bar{\omega} = \sum_{i=0}^{\infty} i \frac{\beta_i}{\beta_{LT}}$$

- It masseurs average reaction time of dependent variable on changes of independent variable
- The smaller the faster reaction

A(R)DL Models

Autoregressive Distributed Lags Model:

- Often it occurs, that by including as an regressors lagged value of dependent variable:
  - better model (in terms of goodness of fit)
  - with smaller number of repressors
  - (with no autocorrelation issue)

  can be derived
- Regressors have to be predetermined (exogenous or lagged endogenous)
- Autocorrelation lead to endogeneity issue → remove of autocorrelation is of high importance
  - Breush – Godfrey test can be used; Durbin – Watson test is inappropriate in this case.
- ARDL model have long-term equilibrium if AR process is stationary

$$y_t = \alpha_1 y_{t-1} + \cdots + \alpha_p y_{t-p} + \mu + \beta_0 x_t + \cdots + \beta_s x_{t-s} + \varepsilon_t$$

Parameter interpretation

ST

$$E(y_t + \Delta y_t) = \alpha_1 y_{t-1} + \cdots + \alpha_p y_{t-p} + \mu + \beta_0(x_t + \Delta x) + \cdots + \beta_s x_{t-s} + \varepsilon_t$$

x_t has no influnece on lagged y as it is in a future

$$E(y_t + \Delta y_t) = E(y_t) + \beta_0 \Delta x$$

$$\boldsymbol{\frac{E(\Delta y)}{\Delta x} = \beta_0}$$

LT

$$E(y_t + \Delta y_t) \\ = \alpha_1 E(y_{t-1} + \Delta y_{t-1}) + \cdots + \alpha_p E\left(y_{t-p} + \Delta y_{t-p}\right) + \mu + \beta_0(x_t + \Delta x) + \cdots \\ + \beta_s(x_{t-s} + \Delta x) + \varepsilon_t$$

$$E(\Delta y_t) = \alpha_1 E(\Delta y_{t-1}) + \cdots + \alpha_p E\left(\Delta y_{t-p}\right) + \Delta x \sum_{i=0}^{p} \beta_i$$

In long-term equilibrium: $E(\Delta y_t) = E(\Delta y_t) = \cdots = E\left(\Delta y_{t-p}\right) = E(\Delta y)$

$$\boldsymbol{\beta_{LT} = \frac{E(\Delta y)}{\Delta x} = \frac{\sum_{i=0}^{\infty} \beta_i}{1 - \alpha_1 - \cdots - \alpha_p}}$$

Diagnostics

- OLS diagnostics (lack of autocorrelation, linear form, heteroscedasticity, normality of residuals, stability)
- Breaush-Godfrey test of no autocorrelation
  - Let's consider a model with autocorrelation of error term:

$$y_t = \mu + \beta_0 x_t + \cdots + \beta_p x_{t-p} + \varepsilon_t$$

$$\varepsilon_t = \rho_1 \varepsilon_{t-1} + \cdots + \rho_p \varepsilon_{t-p} + u_t$$

  - In the first step OLS estimator is fitted in order to obtain $\hat{\varepsilon}_t$, then second regression is estimated:

$$\hat{\varepsilon}_t = \alpha + \alpha_0 x_t + \cdots + \alpha_p x_{t-p} + \rho_1 \hat{\varepsilon}_{t-1} + \cdots + \rho_p \hat{\varepsilon}_{t-p} + u_t$$

  - Joint hypothesis $H0$: $\rho_1 = \cdots = \rho_p = 0$ is tested
  - Test statistics: $(T-p)R_2^2 \sim \chi_p^2$

Choice of number of lags

- F test & Breusch-Godfrey tests
- Lack of autocorrelation testing process
  - Estimation of general model
  - Test on autocorrelation (Breusch-Godfrey test)
    - If autocorrelation exists – new

- Information Citeria & Breusch-Godfrey tests